SOC Alert to Response Simulation Report

VSFTPD Exploitation using Metasploit

1. Introduction

This project demonstrates a complete Security Operations Center (SOC) workflow, including attack simulation, detection, analysis, and response. The attack was performed using Kali Linux as the attacker machine and Metasploitable as the target system.

The objective is to understand how real-world cyberattacks are identified, analyzed, and mitigated using standard SOC practices.

2. Network Configuration

The first step involves identifying the IP addresses of both attacker and target machines to ensure proper communication within the virtual environment.

```
┌──(root㉿kali)-[~]
└─# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:16:ff:97 brd ff:ff:ff:ff:ff:ff
    inet 192.168.19.129/24 brd 192.168.19.255 scope global dynamic eth0
       valid_lft 1777sec preferred_lft 1777sec
    inet6 fe80::20c:29ff:fe16:ff97/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
```

Figure-1: IP address configuration of attacker machine (Kali Linux), showing active network interface and assigned IP used to launch the attack.

```
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:37:03:42
          inet addr:192.168.19.128  Bcast:192.168.19.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fe37:342/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:112 errors:0 dropped:0 overruns:0 frame:0
          TX packets:88 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:8630 (8.4 KB)  TX bytes:10653 (10.4 KB)
          Interrupt:17 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:122 errors:0 dropped:0 overruns:0 frame:0
          TX packets:122 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:33885 (33.0 KB)  TX bytes:33885 (33.0 KB)

msfadmin@metasploitable:~$
```

Figure-2: IP address configuration of target machine (Metasploitable), identifying the vulnerable system used in the exploitation process.

3. Attack Execution

The attack was performed using the Metasploit Framework. A known vulnerability in the VSFTPD service was exploited using the module vsftpd_234_backdoor.

- *RHOST* represents the target machine (Metasploitable)
- *LHOST* represents the attacker machine (Kali Linux)

The exploit was successfully executed, resulting in root-level access to the target system.

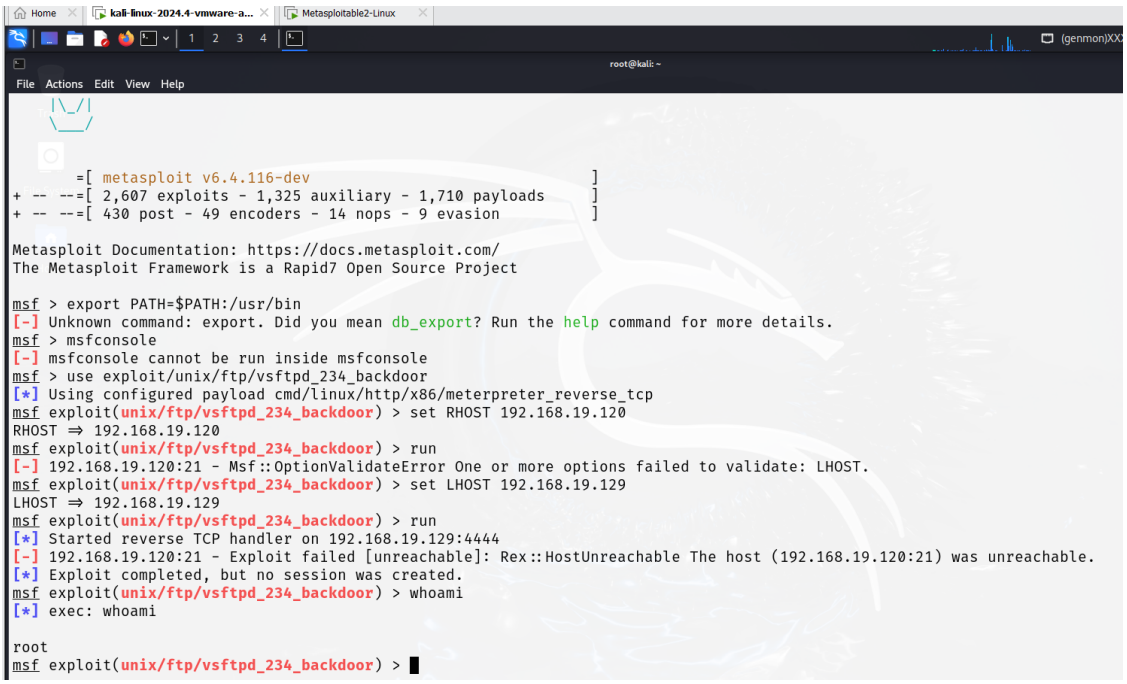

Figure-3: Configuration and execution of the VSFTPD backdoor exploit using Metasploit Framework, demonstrating successful remote access and privilege escalation on the target system.

4. Alert Details

Alert ID	Description	Source IP	Target IP	Priority	Status
001	VSFTPD Exploit	192.168.19.129	192.168.19.120	Critical	open

The alert is classified as Critical due to successful remote exploitation resulting in unauthorized root access. This indicates a severe compromise with high impact on system security.

5. MITRE ATT&CK Mapping

Using MITRE ATT&CK, the attack is mapped to:

- **Technique ID:** *T1190*
- **Technique Name:** Exploit Public-Facing Application

  Explanation: The attacker exploited a vulnerable FTP service (VSFTPD) to gain unauthorized access to the system. This aligns with the MITRE ATT&CK technique T1190, where public-facing applications are targeted.

6. Alert Priority Justification

The alert is classified as Critical because the attacker successfully gained root-level access to the system. This indicates a complete system compromise affecting confidentiality, integrity, and availability.

7. Incident Response Actions

The following actions were taken:

- The affected system was isolated from the network
- The attacker IP was identified and blocked
- Evidence was collected for forensic analysis
- Necessary remediation steps were planned

8. Evidence Collection

Evidence was collected using system-level commands to capture active network connections during the attack.

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > netstat -antp > evidence.txt
[*] exec: netstat -antp > evidence.txt

msf exploit(unix/ftp/vsftpd_234_backdoor) > sha256sum evidence.txt
[*] exec: sha256sum evidence.txt

f64ef28b1f19b7b03549a57a2e57327fc124c5bcfbfffadd9c2cd69f17b94fc0  evidence.txt
msf exploit(unix/ftp/vsftpd_234_backdoor) >
```

Figure-4: Evidence collection and integrity verification using netstat and SHA256 hashing.

Evidence Details:

- Evidence: Network connections log
- Method: netstat command
- Hash Value: *94557aeefaada3c97730020780b3c8c08d40e7ed8fd83c001e8fdd8491c6d2da*

This ensures the integrity of the collected evidence using SHA256 hashing.

9. Incident Timeline

- 11:00 – Attack initiated from Kali Linux
- 11:02 – VSFTPD exploit executed
- 11:03 – Root access gained on target
- 11:05 – Evidence collected
- 11:07 – System isolated

10. Impact Analysis

The attacker gained root-level access to the target system, allowing full control over system operations. This could lead to data breaches, system manipulation, and further network compromise. The impact is considered severe.

11. Recommendations

- Update and patch vulnerable services
- Disable unnecessary services like FTP
- Implement firewall rules
- Deploy intrusion detection systems
- Perform regular security assessments

12. Conclusion

This project successfully demonstrated a full SOC workflow from attack simulation to response. The exploitation of a vulnerable service highlighted the importance of timely patching and continuous monitoring. Proper incident handling and documentation are critical in minimizing security risks.